

Lecture 8: Hedging and Monte Carlo Methods

Delta-Hedging Principles (Seller's Point of View)

1) Selling Call Options and Collecting Premium

When a seller writes (sells) call options, they receive a premium P per option. This premium reflects the risk the seller is taking, and depends on:

- the difference between the current stock price S_0 and the strike price K ,
- the time t left to expiration.

However, the seller faces potential loss if the option ends up in-the-money at expiration:

$$\text{Loss} = S_t - K \quad (\text{if } S_t > K)$$

2) Motivation for Hedging

To offset this potential loss, the seller can purchase a number of shares of the underlying stock at time 0.

Even if the stock increases in value and the seller loses from the short call position, the stock holding gains in value — thus partially offsetting the loss.

Key Insight: If the seller owns the stock from the beginning, even though they sell it below market price at expiration (because of the option), they still gain from the increase in stock price compared to the initial purchase. This helps neutralize the future loss.

3) Delta Hedging at Time 0

At time 0, when selling each call, the seller purchases:

$$\Delta_{C_0} \quad \text{shares of stock per option}$$

where $\Delta_{C_0} = \frac{\partial C(0)}{\partial S(0)}$ is the call option's delta.

At expiration: For each option sold, the total profit is:

$$\text{Profit} = \text{Premium} - C_t + (S_t - S_0) \cdot \Delta_{C_0}$$

- $-C_t$ is the loss from fulfilling the option if exercised.
- $(S_t - S_0)\Delta_{C_0}$ is the gain from holding Δ_{C_0} shares of the stock.
- If $S_t > S_0$, the term $(S_t - S_0)\Delta_{C_0}$ is positive and offsets the loss from $-C_t$.

Conclusion: This hedging strategy reduces the risk the seller faces by dynamically adjusting the stock position to track the option's sensitivity to the underlying.

Remark: Expected Profit with Risk-Free Discounting

To evaluate the profit at time 0 in present value terms, we discount all cash flows using the risk-free rate r .

Total Profit at expiration:

$$\text{Profit} = \text{Premium} - C_t + (S_t - e^{rt}S_0) \Delta_{C_0}$$

The term $(S_t - e^{rt}S_0)$ represents the gain or loss from holding Δ_{C_0} shares of stock over time t , accounting for the risk-free discounting.

Multiple Hedges (Dynamic Delta Hedging)

We now consider the more realistic case where the seller re-hedges at intermediate times. Let:

$$0 < t_1 < t = \text{Expiration}$$

- The seller of the call option receives premium P at time 0.
- At time 0, the seller delta-hedges by buying Δ_{C_0} shares of stock.
- At time t_1 , the seller re-hedges: adjusts the portfolio to now hold $\Delta_{C_{t_1}}$ shares.

Stock profit at time t_1 is:

$$(S_{t_1} - e^{rt_1} S_0) \Delta_{C_0}$$

Discounted back to time 0:

$$\text{Stock Profit at } t_1 = e^{-rt_1} (S_{t_1} - e^{rt_1} S_0) \Delta_{C_0}$$

At expiration: profit from holding the adjusted shares from t_1 to t :

$$\text{Stock Profit at } t = (S_t - e^{r(t-t_1)} S_{t_1}) \Delta_{C_{t_1}}$$

Discounted back to time 0:

$$e^{-rt} (S_t - e^{r(t-t_1)} S_{t_1}) \Delta_{C_{t_1}}$$

初始时刻 $t = 0$:

- 卖方收到 Premium P 。
- 卖方按照期初期权的 Delta 值 Δ_{C_0} 买入股票用于对冲。

到 t_1 : 第一次对冲调整

- 卖方在 t_1 重新评估 delta, 调整持仓至 $\Delta_{C_{t_1}}$ 股。
- 这意味着, 从 $t = 0$ 到 t_1 期间, 卖方持有 Δ_{C_0} 股的收益如下:

$$\text{Stock profit at } t_1 = (S_{t_1} - e^{rt_1} S_0) \cdot \Delta_{C_0}$$

- S_{t_1} : 当时市场上的股票价格;
- $e^{rt_1} S_0$: 这部分股票若存入无风险银行, 到 t_1 应有的价值;
- 所以这两个差值衡量了股票对风险中性账户的**超额收益**。

贴现到时间 $t = 0$:

$$e^{-rt_1} (S_{t_1} - e^{rt_1} S_0) \cdot \Delta_{C_0}$$

从 t_1 到 t : 第二段持股期

- 现在你持有的是 $\Delta_{C_{t_1}}$ 股。
- 同样地，到期时股票变动的利润为：

$$\text{Stock profit at } t = (S_t - e^{r(t-t_1)} S_{t_1}) \cdot \Delta_{C_{t_1}}$$

贴现回时间 0:

$$e^{-rt} (S_t - e^{r(t-t_1)} S_{t_1}) \cdot \Delta_{C_{t_1}}$$

到期时，作为卖方，还要支付期权赔偿

- 如果 $S_t > K$ ，你必须以 K 卖出股票，所以亏了 $S_t - K$ ；
- 期权 payoff 是：

$$\max(S_t - K, 0)$$

贴现到时间 0:

$$e^{-rt} \max(S_t - K, 0)$$

注意：这里是负现金流（你要赔钱），所以是带负号的。

Total Profit at Time 0 (Discounted)

Combining all the components:

$$\text{Total Profit (Time 0)} = e^{-rt} \cdot \max(S_t - K, 0) + e^{-rt_1} (S_{t_1} - e^{rt_1} S_0) \Delta_{C_0} + e^{-rt} (S_t - e^{r(t-t_1)} S_{t_1}) \Delta_{C_{t_1}}$$

- The first term is the payoff of the short call (negative cash flow to seller).
- The second term is the profit from holding Δ_{C_0} shares from 0 to t_1 .
- The third term is the profit from holding $\Delta_{C_{t_1}}$ shares from t_1 to t .

This model generalizes easily to multiple rebalancing times: $t_1, t_2, \dots, t_n = t$.

General Case: Delta-Hedging at Multiple Time Steps

More generally, suppose the seller rebalances the portfolio to maintain delta-neutrality at a sequence of times:

$$0 = t_0 < t_1 < t_2 < \dots < t_n = t$$

At each time t_i , the seller adjusts the number of shares held to match the updated delta $\Delta_{C_{t_i}}$.

The total profit and loss (P&L), discounted to time 0, is:

$$\text{P\&L}_{\text{seller}} = \text{Premium} - e^{-rt} C_t + \sum_{i=1}^n e^{-rt_i} (S_{t_i} - e^{r(t_i-t_{i-1})} S_{t_{i-1}}) \Delta_{C_{t_{i-1}}}$$

Explanation:

- The first term is the premium received from selling the call option.
- The second term $e^{-rt}C_t$ is the (discounted) option payoff that must be paid if the option is exercised.
- The third term is the sum of profits from holding shares of the stock between rebalancing times t_{i-1} and t_i , using the previously chosen delta $\Delta_{C_{t_{i-1}}}$.

Each stock profit term is discounted back to time 0 using the factor e^{-rt_i} , and reflects the profit due to price movement in S over $[t_{i-1}, t_i]$.

This formula forms the foundation for Monte Carlo simulation of delta-hedging profit under discrete rebalancing frequency.

基础例子

- 当前时间 $t = 0$
- 当前股票价格 $S_0 = 100$
- 执行价格 $K = 105$
- 卖出一个看涨期权，收取 Premium = \$5
- 假设 Delta 值 $\Delta_{C_0} = 0.5$ ，所以你买入 0.5 股股票对冲

到期时的两种情况 ($t = T$)

情况 1: 股价上涨到 $S_T = 110$

计算卖方亏损 (因为买方会行权) :

买方以 $K = 105$ 向你买入股票。市场价是 $S_T = 110$ ，你必须以低价卖出股票，亏损:

$$\text{Option Loss} = S_T - K = 110 - 105 = 5$$

你持有的 0.5 股票升值:

股票从 100 涨到 110，每股涨了 10，所以你赚到:

$$\text{Stock Profit} = \Delta_{C_0} \cdot (S_T - S_0) = 0.5 \cdot (110 - 100) = 5$$

计算总利润:

$$\text{Total Profit} = \text{Premium} - \text{Option Loss} + \text{Stock Profit} = 5 - 5 + 5 = 5$$

对冲成功!

情况 2: 股价下跌到 $S_T = 95$

买方不行权:

因为 $S_T = 95 < K = 105$, 期权没有价值, 你无需支付赔付:

$$\text{Option Loss} = 0$$

你持有的 0.5 股票亏损:

股票从 100 跌到 95, 每股跌了 5, 你亏:

$$\text{Stock Loss} = \Delta_{C_0} \cdot (S_T - S_0) = 0.5 \cdot (95 - 100) = 0.5 \cdot (-5) = -2.5$$

计算总利润:

$$\text{Total Profit} = \text{Premium} - \text{Option Loss} + \text{Stock Profit} = 5 - 0 + (-2.5) = 2.5$$

虽然股票亏损, 但你依旧盈利, 只是少了一些。

总结对冲策略下的利润计算公式

对于任意 S_T , 总利润为:

$$\text{Profit} = P - \max(S_T - K, 0) + \Delta_{C_0}(S_T - S_0)$$

其中:

- P 表示权利金;
- $\max(S_T - K, 0)$ 是你必须赔付的金额 (当期权被行权时);
- $\Delta_{C_0}(S_T - S_0)$ 是你持有的股票因股价波动产生的对冲收益。

在 Monte Carlo 模拟中，我们通常用样本均值：

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

来估计一个期望值 $\mathbb{E}[X]$ 。由于模拟是随机的，每次模拟的结果会有波动，这种波动的大小用标准误差衡量：

$$\sigma_{\bar{X}} = \frac{\sigma_n}{\sqrt{n}}$$

而根据中心极限定理，当 n 很大时，样本均值服从正态分布，因此：

区间宽度	覆盖度
$\pm 1\sigma$	68%
$\pm 2\sigma$	95%
$\pm 3\sigma$	99.7%

计算期望收益 (Expected Profit) :

在 100,000 次模拟中你购买期权后的平均收益，即：

$$\mathbb{E}[\text{Profit}] = \frac{1}{n} \sum_{i=1}^n \text{Profit}_i$$

计算标准误差 (Standard Error) :

计算样本均值的标准误差：

$$\text{SE} = \frac{\sigma}{\sqrt{n}}$$

这是用于衡量你的 Monte Carlo 均值估计的可靠性。“我这个模拟均值与真实期望值之间的典型误差是多少？”